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END SEMESTER EXAMINATION- DECEMBER 2025

Name of the Course: B. Tech

Name of the Paper : Advanced Structural Analysis

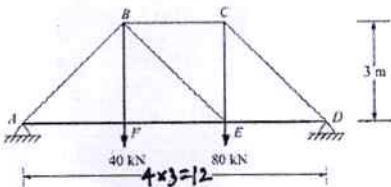
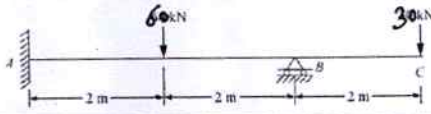
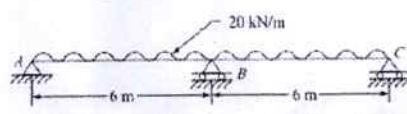
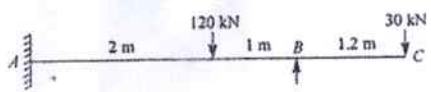
Time: 3 Hours

Semester : Vth Semester

Paper Code : TCE 501

Maximum Marks : 100

- Note: i) All questions are compulsory
 ii) Answer any two questions from each sub question.
 iii) Each question carries 10 Marks

Q.1	Solve any Two	(20 Marks)	
a)	A two hinged parabolic arch of span 20 m and rise 4m carries a uniformly distributed load of 50kN/m on the left half of the span. Find the reactions at the supports and the position and amount of maximum bending moment.	(CO 1)	
b)	The three-hinged stiffening girder of a suspension bridge of span 120 m is subjected to two point loads of 240 kN and 300 kN at distances 25 m and 80 m from the left end. Find the shear force and bending moment for the girder at a distance of 40 m from the left end. The supporting cable has a central dip of 12 m. Find also the maximum tension in the cable, and draw the B.M. diagram for the girder.	(CO 1)	
c)	A suspension bridge of span 80 m and width 6 m is having two cables stiffened with two hinged girders. The central dip of cables is 8 m, the dead load on the bridge is 5 kN/m ² and the live load is 10 kN/m ² which covers the left-half of the span. Determine the shear force and bending moment at 20 m from the left end. Find also the maximum tension in the cable.	(CO 1)	
Q.2	Solve any Two	(20 Marks)	
a)	Determine the force in the members of the truss shown in Fig. The cross-sectional area of vertical and horizontal members is 4000 mm ² and that of the diagonals is 6000 mm ² .	(CO 2)	
			
b)	Determine the reaction components in the beam shown in Fig. by consistent deformation method. EI is constant throughout.	(CO 2)	
			
c)	Determine the reaction components in the beam shown in Fig. Assume flexural rigidity to be the same throughout.	(CO 2)	
			
Q.3	Solve any Two	(20 Marks)	
a)	Determine the support moments and draw the B.M. diagram by moment distribution method of the following beam.	(CO 3)	
			

	b)	Analyze the beam as shown in Fig. by slope deflection method.	(CO 3)
	c)	Analyze the portal frame as shown in Fig. by moment distribution method.	(CO 3)
Q.4		Solve any Two (20 Marks)	
	a)	Analyze the following beam by stiffness matrix method.	(CO 4)
	b)	Analyze the following portal frame by stiffness matrix method.	(CO 4)
	c)	Analyze the following beam by stiffness matrix method.	(CO 4)
Q.5		Solve any Two (20 Marks)	
	a)	Write the assumptions of plastic theory (at least 10).	(CO 5)
	b)	Cross section of the beam as shown in Fig. is subjected to sagging bending moment. Find the shape factor if permissible yield stress in compression and tension is 200 MPa and 240 MPa respectively.	(CO 5)
	c)	Find the collapse load for a fixed beam carrying an U.D.L of W_u kN/m through the span of "L" m.	(CO 5)